

1D CNN — Architecture Variant Comparison

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Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Total trials	325 (train pool 292, hold-out 33)
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	20 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	per_trial_zscore
CV folds · shuffle	7 · True
Augmentation	on · copies=6 · time_shift, time_warp, time_mask, amplitude_scale, baseline_offset, channel_dropout
Epochs · batch size	60 · 16

Sensor grid (✓ enabled, ✗ disabled, — not present)

Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✗	✗	—	—
left thumb	✗	✗	✓	✓
left index	✗	✗	✓	✓
left middle	✗	✗	✓	✓
left ring	✗	✗	✓	✓
left pinky	✗	✗	✓	✓
left wrist	✗	—	—	—
right palm	✗	✗	—	—
right thumb	✗	✗	✓	✓
right index	✗	✗	✓	✓
right middle	✗	✗	✓	✓
right ring	✗	✗	✓	✓
right pinky	✗	✗	✓	✓
right wrist	✗	—	—	—

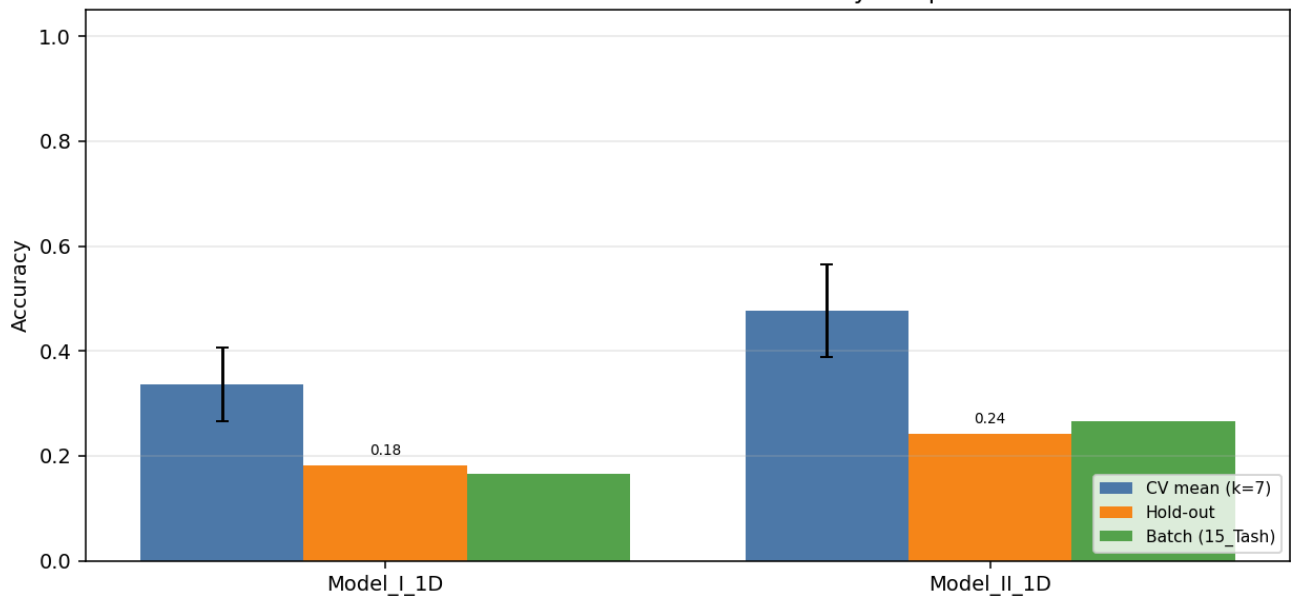
Enabled: 20 · Disabled: 26 · Modalities: Flex → 20 channels total.

Variant comparison (summary)

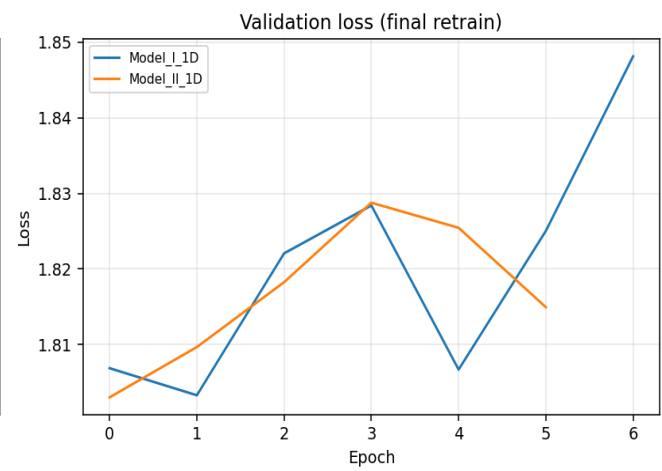
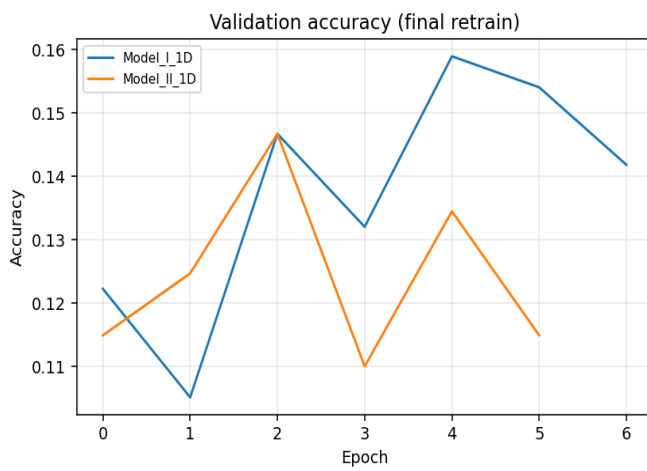
Variant	Params	CV mean (k=7)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Model_I_1D	118,630	0.3355	0.0703	0.1818	1.7729	0.1667	5/30
Model_II_1D	73,926	0.4765	0.0888	0.2424	1.7799	0.2667	8/30

Best on hold-out: **Model_II_1D** · Best on CV: **Model_II_1D**

1D CNN architecture variants — accuracy comparison



Training curves (final retrain on full training pool)



Variant: Model_I_1D

1-D adaptation of published Model I:
Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax

Trainable params	118,630
CV mean ± std	0.3355 ± 0.0703
Hold-out test acc / loss	0.1818 · 1.7729
Batch-test acc	0.1667 (5/30)

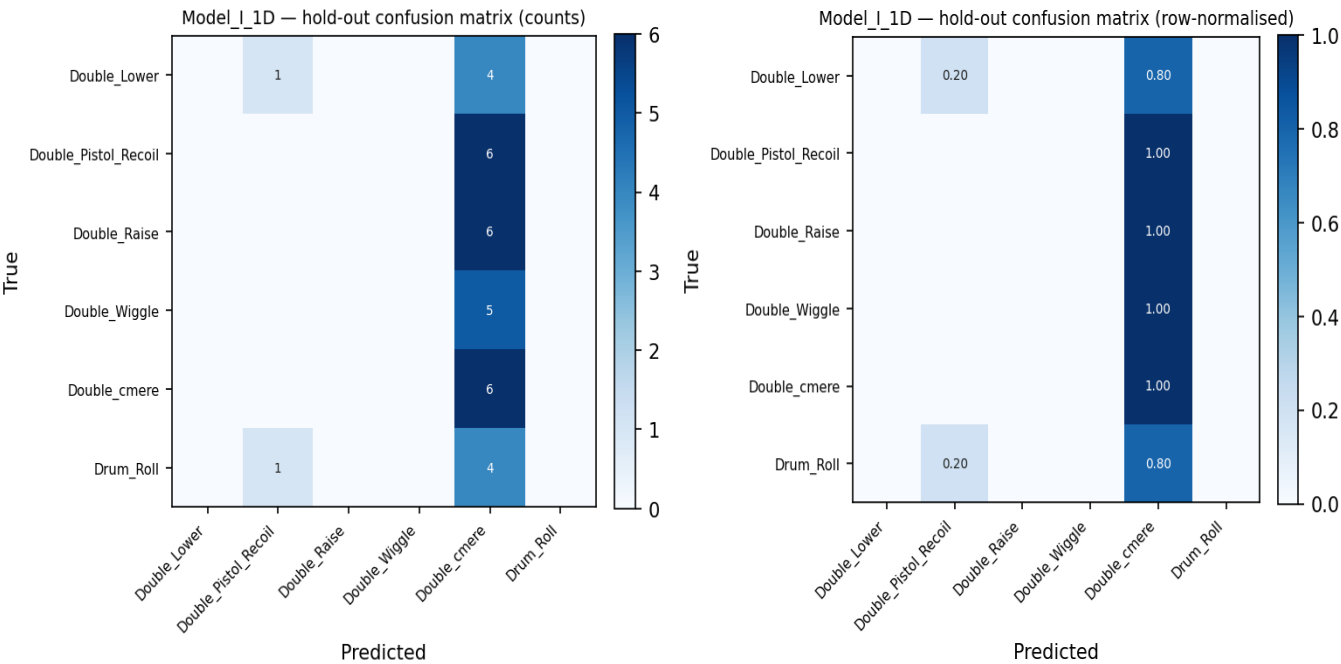
Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1750	42	0.2857	0.3333	0.3333
2	1750	42	0.3095	0.4524	0.4524
3	1750	42	0.3333	0.3571	0.3333
4	1750	42	0.2857	0.2857	0.2381
5	1750	42	0.5000	0.5476	0.5238
6	1757	41	0.2927	0.3415	0.2683
7	1757	41	0.3415	0.3902	0.3415

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.000	0.000	0.000	5
Double_Pistol_Recoil	0.000	0.000	0.000	6
Double_Raise	0.000	0.000	0.000	6
Double_Wiggle	0.000	0.000	0.000	5
Double_cmere	0.194	1.000	0.324	6
Drum_Roll	0.000	0.000	0.000	5
macro avg	0.032	0.167	0.054	33
weighted avg	0.035	0.182	0.059	33

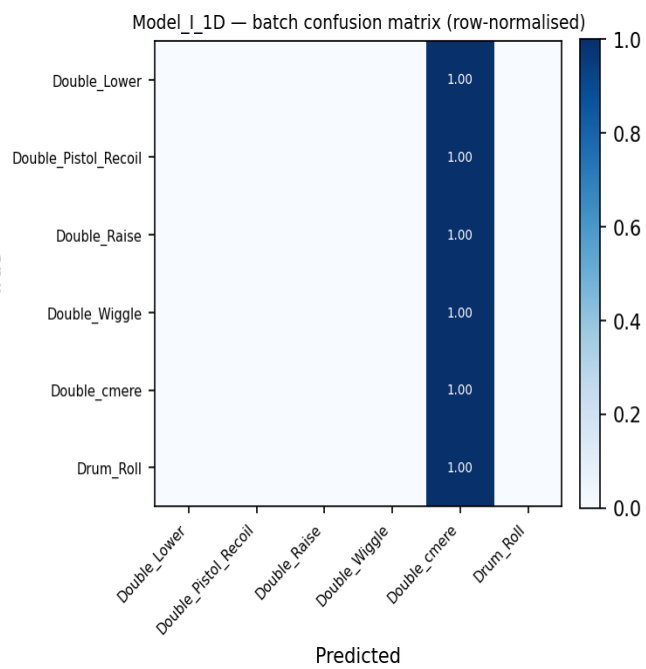
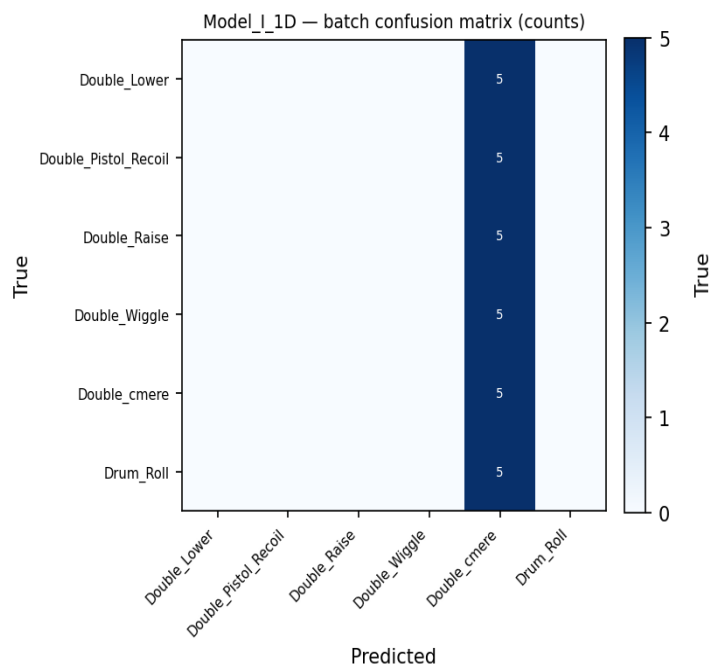
Hold-out confusion matrix



Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Pistol_Recoil	0	5	0.000
Double_Raise	0	5	0.000
Double_Wiggle	0	5	0.000
Double_cmere	5	5	1.000
Drum_Roll	0	5	0.000
Overall	5	30	0.167

Batch confusion matrix (15_Tash)



Variant: Model_II_1D

1-D adaptation of published Model II (CNN+LSTM):
Conv1D(32,k=5,s=3)→Conv1D(32,k=3)→BN→Pool→Drop→LSTM(64,seq)→Drop→LSTM(64)→Drop→Dense(128)→BN→Drop→Softmax

Trainable params	73,926
CV mean ± std	0.4765 ± 0.0888
Hold-out test acc / loss	0.2424 · 1.7799
Batch-test acc	0.2667 (8/30)

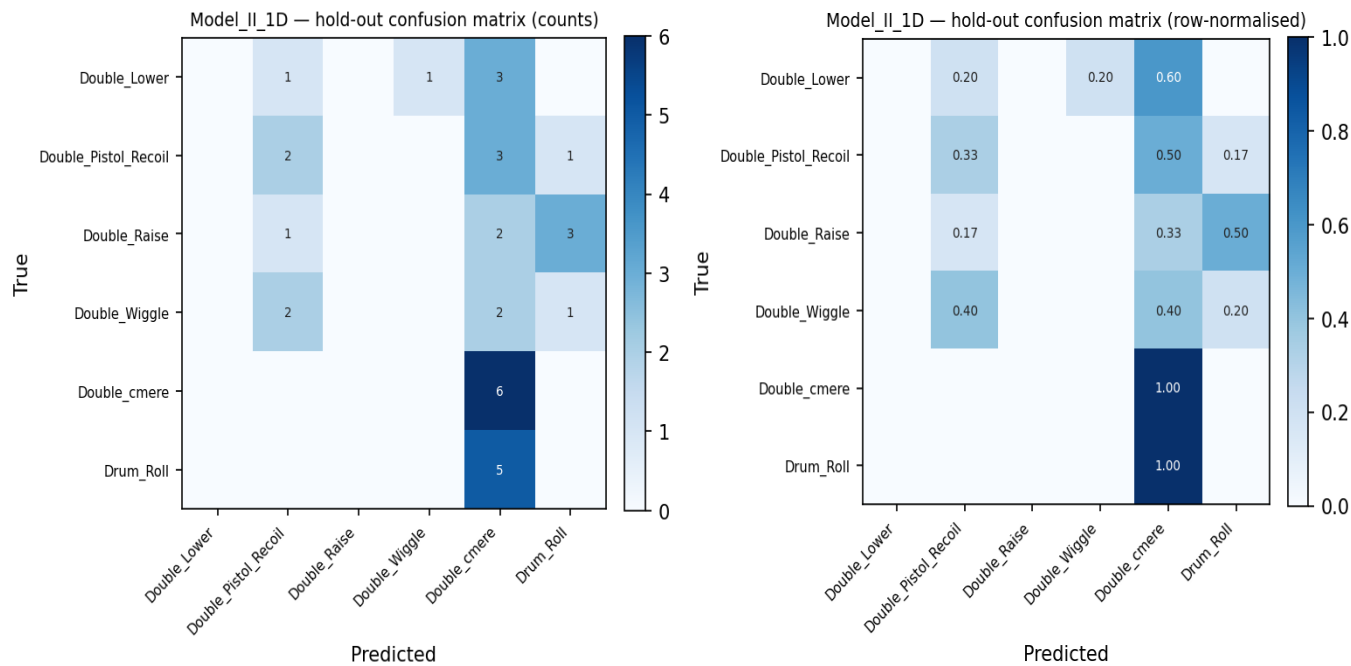
Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1750	42	0.3571	0.4286	0.2857
2	1750	42	0.5714	0.6190	0.5714
3	1750	42	0.3571	0.4762	0.3810
4	1750	42	0.4286	0.5000	0.4762
5	1750	42	0.5238	0.5714	0.5000
6	1757	41	0.5122	0.5122	0.4634
7	1757	41	0.5854	0.6098	0.5122

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.000	0.000	0.000	5
Double_Pistol_Recoil	0.333	0.333	0.333	6
Double_Raise	0.000	0.000	0.000	6
Double_Wiggle	0.000	0.000	0.000	5
Double_cmere	0.286	1.000	0.444	6
Drum_Roll	0.000	0.000	0.000	5
macro avg	0.103	0.222	0.130	33
weighted avg	0.113	0.242	0.141	33

Hold-out confusion matrix



Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Pistol_Recoil	2	5	0.400
Double_Raise	0	5	0.000
Double_Wiggle	0	5	0.000
Double_cmere	4	5	0.800
Drum_Roll	2	5	0.400
Overall	8	30	0.267

Batch confusion matrix (15_Tash)

